

Professional Summary

Information Science undergraduate with a 9.1/10 CGPA and hands-on experience in AI/ML, data engineering, and application development. Skilled in Python, PyTorch, Scikit-learn, REST APIs, and predictive analytics, with experience in GNNs, knowledge graphs, and ML forecasting. SAP Certified Backend Developer with experience in Docker, AWS, Google Cloud, and Git.

Education

B M S Institute of Technology and Management, Bengaluru

B.E. in Information Science, CGPA: 9.1/10

K M W A PU College

Grade XII – PCMB, 94.67%

Vijayashree Public School

Grade X – ICSE, 92.33%

Projects

NeuroGraph – BioAI Drug Repurposing Platform

[GitHub](#) [Live Website](#)

- Built a neuro-symbolic AI platform combining Graph Neural Networks (GNNs) with knowledge-graph reasoning to identify and rank potential drug-repurposing candidates by analyzing relationships across drugs, diseases, genes, pathways, and treatments.
- Designed an explainable prediction pipeline using GNN-based scoring and rule-based reasoning to strengthen recommendations, uncover meaningful biological relationships, and provide interpretable insights from the connected biomedical knowledge graph.
- Developed a PyTorch-based chest X-ray analysis module using a pre-trained DenseNet121 model to detect abnormalities and map visual findings to relevant diseases, adding an additional deep-learning-based diagnostic layer to the drug-repurposing platform.
- Tools: Python, FastAPI, PyTorch, DenseNet121, Neo4j Aura, React JS, TypeScript, D3.js

PERM – Predictive Edge Resource Management using ML

- Built a predictive edge-resource management system that monitors CPU utilization, memory consumption, and response latency under simulated workloads to anticipate performance issues and maintain reliable application performance.
- Developed a forecasting pipeline using a VotingRegressor ensemble to predict CPU utilization and response latency 30 seconds ahead, using sliding-window telemetry features and time-based data splitting for reliable predictions.
- Implemented proactive SLO-based alerting and automated remediation to identify predicted resource breaches early, generate warnings, and dynamically adjust system capacity based on forecasted workload and recovery conditions.
- Tools: Python, Flask, REST APIs, Scikit-learn, Locust, Psutil, Pandas, NumPy, Git

Decentralized Android Bluetooth Chat Application

- Built an Android decentralized chat application using Bluetooth Low Energy (BLE) Mesh for offline, multi-hop communication between nearby smartphones without internet, cellular networks, or external servers.
- Implemented multi-hop forwarding with TTL-based routing, node discovery, and duplicate-message filtering to relay text reliably as users move between network nodes.
- Integrated real-time GPS sharing and SOS broadcasts using compact BLE packets, relaying priority alerts and location updates across the mesh where conventional connectivity is unavailable.
- Tools: Java, Android, Bluetooth Low Energy Mesh, Bluetooth GATT, GPS, Git

Technical Skills

Languages: Python, C++, C, R

Frameworks & Web: React JS, Flask, FastAPI, REST APIs, HTML, CSS, Bootstrap

Data & Visualization: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Power BI, D3.js, Matplotlib

Databases: SQL, MySQL, Neo4j Aura, Graph Database, Supabase

Cloud & DevOps: Docker, AWS, Google Cloud, SonarQube, Git, GitHub

Engineering: Software Testing, Debugging, Data Analysis, Application Integration, Automation

Tools: Jira, Locust, Psutil, VS Code, Android Studio, Bluetooth Mesh

Certifications

- SAP Certified - Backend Developer - SAP Cloud Application Programming Model
- Google Cloud Foundations Certificate - Badge
- AWS Cloud Practitioner Essentials - Digital Course
- ServiceNow Administration Fundamentals - Badge

Achievements

- **CODE-RED 2025 – Unisys Track:** Ranked among the Top 10 teams in a national-level 24-hour hackathon held at BMSITM.
- **1st Place, SenseCrypt:** Secured 1st place in an intra-college hackathon organized by the IEEE-EMBS Society, BMSITM.